

// Day 26 of 30

COMPARISON

RabbitMQ vs Kafka — Choose Wisely

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Core Difference

Message broker vs event stream

RabbitMQ = Smart Broker

- Routes messages to consumers
- Message deleted after consumed
- Traditional messaging

Kafka = Dumb Broker, Smart Consumer

- Append-only log
- Messages persist (configurable)
- Consumers track their position

💡 RabbitMQ delivers. Kafka stores.



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When RabbitMQ Wins

Task queues, routing, simple async

Best for:

- Send email after signup
- Process payment async
- Route to specific consumer
- Request-reply pattern
- Simple pub-sub

Examples:

- Notification service
- Background job processing
- Order → kitchen routing

💡 RabbitMQ = simple, reliable, easy to set up



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When Kafka Wins

High throughput, event sourcing, streaming

Best for:

- Millions of events/second
- Event sourcing (replay events)
- Stream processing
- Multiple consumers same data
- Analytics pipeline

Examples:

- User activity tracking
- Real-time analytics
- Log aggregation

💡 Kafka = massive scale, event history, streaming



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Performance

Numbers comparison

Throughput:

RabbitMQ: ~50,000 msg/sec
Kafka: ~1,000,000 msg/sec

Latency:

RabbitMQ: ~1ms (lower)
Kafka: ~5ms (higher)

Message retention:

RabbitMQ: deleted after consume
Kafka: kept for days/weeks

💡 Need speed? Kafka. Need low latency? RabbitMQ.



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Complexity

Setup and maintenance

RabbitMQ:

- Easy to set up (5 minutes)
- Web management UI built-in
- Good documentation
- Single binary

Kafka:

- Needs Zookeeper (or KRaft)
- More ops knowledge needed
- Partition management
- Steeper learning curve

💡 Small team? Start with RabbitMQ. Scale? Add Kafka later.



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The Decision Framework

Simple flowchart

Need replay of old events?

→ Kafka

Need complex routing?

→ RabbitMQ

Millions of events/second?

→ Kafka

Simple async tasks?

→ RabbitMQ

Multiple consumers same event?

→ Kafka

First time using message queues?

→ RabbitMQ

💡 Most backend apps start with RabbitMQ. That's fine.



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